

KREITIKA YADAV

kreitikayadav@gmail.com · +91-8800776380 · [LinkedIn](#) · [GitHub](#) · M.Sc. ML & Signal Processing, NTU Singapore (Aug 2026) · GATE CS 98.9%ile (AIR 1951)

EDUCATION

M.Sc. Machine Learning & Signal Processing · [Nanyang Technological University, Singapore](#) Aug 2026 – 2027

B.Tech Electrical Engineering (Minor: CS) · [Delhi Technological University](#) · CGPA 7.99/10 May 2021 – May 2025

TECHNICAL SKILLS

ML / NLP / DL: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, LlamaIndex, Scikit-learn, OpenCV, SHAP

Data Engineering: dbt, Apache Spark, Airflow, Snowflake, AWS (S3, Glue, Redshift), BigQuery, PostgreSQL, Redis

Languages & Tools: Python, C++, SQL, Bash · Docker, MLflow, FastAPI, Weights & Biases, Great Expectations

PROJECTS

RAG-Powered Financial Intelligence Platform · [LangChain](#) · [Pinecone](#) · [OpenAI API](#) · [FastAPI](#) · [PostgreSQL](#) Apr 2026 – Present

- Architected a production-grade RAG system ingesting 10,000+ SEC filings and earnings transcripts; implemented hybrid BM25 + dense vector search over Pinecone with custom metadata filters (ticker, sector, filing date)
- Reduced hallucination rate by 34% (RAGAS faithfulness) vs naive chunking via adaptive sliding-window chunking; served via FastAPI with sub-200ms p95 latency; tracked context recall 0.91, answer relevancy 0.88 on 500 curated QA pairs

Multilingual Sentiment & Stance Detection — Hinglish Code-Mixed Text · [MuRIL](#) · [XLM-RoBERTa](#) · [LoRA/PEFT](#) · [HuggingFace](#) · [WandB](#) Feb 2026 – Apr 2026

- Fine-tuned MuRIL and XLM-RoBERTa-large on 18,000 Hinglish social-media posts for sentiment & stance classification; applied LoRA (rank-16) reducing trainable params by 87%, achieving F1 0.83/0.79 — outperforming full fine-tune baseline by 2.1 F1 points
- Boosted minority-class recall by 11pp via back-translation augmentation (Helsinki-NLP); tracked 90+ experiments in W&B; model cards and dataset hosted on HuggingFace Hub with 200+ downloads

LLM SQL Generator & Auto-Documentation Engine · [LlamaIndex](#) · [GPT-4o](#) · [dbt](#) · [SQLAlchemy](#) · [Streamlit](#) · [Redis](#) Dec 2025 – Feb 2026

- Built text-to-SQL system with schema-aware prompting over 120-table warehouse schema; achieved 78% execution accuracy on Spider benchmark and 91% on internal queries; schema-linking pruned irrelevant context by 60%, cutting token cost ~40%
- Automated dbt model documentation for 300+ models via GPT-4o structured output; deployed as Streamlit app with Redis caching and RBAC, processing 1,000+ analyst queries in beta with 94% positive feedback

Abstractive Clinical Note Summarization with FLAN-T5 · [FLAN-T5](#) · [HuggingFace](#) · [RLHF-lite](#) · [MIMIC-III](#) · [BERTScore](#) Oct 2025 – Dec 2025

- Fine-tuned FLAN-T5-Large on 15,000 de-identified MIMIC-III notes; achieved ROUGE-L 0.42 and BERTScore F1 0.88, surpassing BART and Pegasus baselines by 3–5 ROUGE points; applied RLHF-lite with NLI-based reward model to control hallucinations

WORK EXPERIENCE

Business Analyst – Data & Analytics · [TriGen Wealth](#) · [Mumbai](#) Jan 2026 – Present

- Architected end-to-end cloud data warehouse on AWS (S3 → Glue ETL → Redshift), consolidating 8 siloed sources (Bloomberg feeds, custodian files, CRM, trading logs) into a single source of truth for 25+ analysts and PMs
- Designed star-schema model (3 fact tables, 12 dimension tables) and dbt transformation layer with 60+ models and 200+ data quality tests; built Airflow DAGs reducing data-availability SLA from T+2 to T+1
- Developed VaR, Sharpe, drawdown, and factor-exposure risk modules processing 5M+ daily NAV records; Great Expectations monitoring caught 15+ critical upstream issues/month, saving ~20 analyst-hours weekly

Software Engineering Intern · [Byte2Bit \(Aarhus University Startup\)](#) · [Remote](#) Jul 2025 – Dec 2025

- Designed and implemented a distributed AIS data pipeline in C++ processing 200M+ maritime records, with custom hash-map indexing for O(1) MMSI lookups, spatial bounding-box filters, and lock-free concurrent reads — reducing average query latency by 45%
- Built a compression benchmarking framework comparing FSST, LZ4, Snappy, and Zstd on a 25GB AIS dataset; integrated SIMD-vectorized encode/decode loops and multi-threaded chunk schedulers, achieving >30% latency reduction and 1.4× faster decompression throughput
- Developed Python analytics layer over the C++ engine exposing a Pandas-compatible API; ran exploratory vessel trajectory analysis and anomaly flagging (speed violations, AIS spoofing) used directly in client-facing dashboards

Software Engineer Intern – AI & Verification · [Siemens EDA](#) · [Noida](#) Jun 2025 – Aug 2025

- Built a recursive information extraction pipeline (outperforming standard RAG across 15+ use cases) to parse dense hardware specification documents and auto-generate 10,000+ lines of SystemVerilog and protocol tree stubs, accelerating the verification lifecycle end-to-end
- Trained a transformer-based anomaly detection model on 50K+ protocol transaction logs to flag timing violations and protocol deviations with exact timestamp precision; integrated alerts into the CI regression pipeline
- Automated constrained-random stimulus generation and SVA assertion authoring via LLM-assisted tooling, reducing manual verification authoring time by 40% across AMBA AHB/APB subsystem bring-up

DL Research Intern · **Control Systems Lab, DTU** · Delhi

Sep 2024 – Apr 2025

- U-Net segmentation pipeline on 1,000+ PPMI MRI volumes for Parkinson's Disease; achieved 25% Dice improvement (0.68 → 0.85); co-authored ICAISI-accepted paper with Dr. Sudarshan Valluru

ML Research Intern · **National University of Singapore (NUS)** · Singapore

Dec 2022

- Regression models on 10k+ telemetry samples (+15% accuracy); OCR-based ALPR pipeline with 90%+ recognition on 500+ plates; awarded HPE Letter of Recommendation for technical excellence

ACHIEVEMENTS & ACTIVITIES

GATE CS 2025: 98.9%ile, AIR 1951 (top ~1% among 200k+ candidates) · **IEEE WIE Lead, DTU:** Organized 50+ technical events, led flagship INVICTUS (2022–2025) · **PQC Research:** Co-authored survey on 14 IoT PQC integration strategies (NIST/OQS/ETSI aligned) from review of 90+ papers